

Golden Rule -> evaluate retrieval and generation separately - a good answer can hide a bad retriever, and vice versa

1. Why Evaluate RAG

A RAG system has two places to fail: retrieval (did we find the right chunks?) and generation (did the LLM use them correctly?). Judging only the final answer can't tell you which one is broken — you have to check both.

2. Retrieval Metrics

Metric	What It Measures	Notes
Precision@k	Of the k chunks retrieved, how many are relevant	Higher = less noise in retrieved context
Recall@k	Of all relevant chunks that exist, how many were retrieved	Higher = fewer relevant chunks missed
MRR (Mean Reciprocal Rank)	How high the first relevant chunk ranks	Rewards relevant results appearing early

3. Generation Metrics

Metric	What It Measures
Faithfulness / groundedness	Does the answer only state things supported by the retrieved context?
Answer relevancy	Does the answer actually address the question that was asked?
Context relevancy	Was the retrieved context itself actually relevant to the question?

4. Tools

- **RAGAS** — open-source framework for automated RAG evaluation (faithfulness, relevancy, and more) using an LLM-as-judge.
- **LangSmith** — tracing and evaluation platform, useful for debugging chains and running evaluation datasets.
- **Manual spot-checks** — still essential; automated metrics don't catch everything, especially domain-specific correctness.

5. Manual Evaluation Checklist

- Does the retrieved context actually contain the answer?
- Does the generated answer match the retrieved context, with no hallucinated extras?
- Are citations / sources correct?
- Does the system say “I don't know” when the answer isn't in the knowledge base?

Pitfall

A RAG system can produce a fluent, confident-sounding answer that is completely unsupported by the retrieved context. Always check faithfulness, not just fluency.

Remember

Build a small evaluation set of representative questions (with known correct answers and sources) early, and rerun it whenever you change chunking, embeddings, or prompts.

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INTERVIEW PREP

Q: Why evaluate retrieval and generation separately instead of just checking the final answer?

A: A wrong final answer could stem from bad retrieval (the right context was never found) or bad generation (the LLM ignored good context) — separating the two pinpoints where to fix the pipeline.

Q: What does “faithfulness” measure in RAG evaluation?

A: Whether every claim in the generated answer is actually supported by the retrieved context, i.e. the model isn't adding unsupported or hallucinated information.

Q: What's the difference between precision@k and recall@k for a retriever?

A: Precision@k measures how many of the k retrieved chunks are actually relevant; recall@k measures how many of all relevant chunks in the corpus were successfully retrieved.

Q: Why is a manual evaluation set still necessary even with tools like RAGAS?

A: Automated LLM-as-judge metrics can miss domain-specific correctness issues and can themselves be wrong; a curated set of real questions with known answers grounds evaluation in ground truth.

Q: What should a well-behaved RAG system do when the answer isn't in the knowledge base?

A: Say it doesn't know, or that it can't find the answer in the provided documents, rather than guessing or falling back to unguided LLM knowledge.